MILITARY
SPECIFICATION

MIL-A-10894C(SigC)
30 June 1959
Superseding
MIL-A-10894B(SigC)
18 July 1958

AMPLIFIER-POWER SUPPLY AM-598( )/U

1. SCOPE

1.1 This specification covers one type of audio amplifier and a vibrator power supply housed in a common enclosure, designated as Amplifier-Power Supply AM-598( )/U. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping.
QQ-S-571	Solder; Lead Alloy; Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form.
UU-T-111	Tape; Paper, Gummed (Sealing and Securing).
DDD-B-20	Bags, Mailing (Cotton).
LLL-F-291	Fiberboard; Corrugated, Single-face (Flexible.)
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiber.
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing.

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MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior, Packaging.
MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-S-901	Shockproof Equipment, Class H-I (High Impact), Shipboard.
MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

FSC 5820

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STANDARDS

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MIL-STD-22	Welding; Joint Designs, Data for (Structures Other Than Armor).
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-122	Color Code for Chassis Wiring for Electronic Equipment.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-B-61578	Chassis Grounding for Electronic Equipment.
SC-DL-104264	Amplifier Power Supply, Part of AM-598( )/U.
SC-DL-104390	Mounting MT-759( )/PRC.
SC-DL-104397	Cable Assembly CX-2031( )/U.
SC-DL-104411	Case, Accessories CY-1210( )/U.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification No. 4.

Consolidated Freight Classification Rules No. 21.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification No. A-3 and No. 14.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth St. NW Washington 6, D. C.)

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3. REQUIREMENTS

3.1 Construction.- Amplifier-Power Supply AM-598( )/U consisting of the following units shall be constructed in accordance with the listed drawings:

Amplifier Power Supply, Part of AM-598( )/U	SC-DL-104264
Mounting MT-759( )/PRC	SC-DL-104390
Cable Assembly CX-2031( )/U	SC-DL-104397
Case, Accessories CY-1210( )/U	SC-DL-104411

3.2 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.2.1 Slack.- Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.2.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields and similar items), they shall be protected by suitable grommets or bushings.

3.2.3 Support.- Wires and cables shall be properly supported and secured, to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.2.4 Clearance.- Clearance between wires or cables, and parts such as electron tubes and resistors, shall be sufficient to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, or other parts, shall be at least 1/4 inch unless use of specified parts makes such clearance impracticable.

3.2.5 Splicing.- Wires in a continuous run between two terminals shall not be spliced during assembly of equipment.

3.2.6 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. Except during overall tropicalization of the equipment, no varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.3 Grounding.- Ground connections to shields and to other mechanical parts, except to chassis or frame, shall not be made to complete electrical circuits but only to eliminate high-potential a.c. points.

3.3.1 Grounding to chassis.- Grounding to chassis shall conform to Drawing SC-B-61578

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3.3.2 Shielding on wire and cable.- Shielding on wire or cable shall be grounded to the chassis or frame, in the manner specified in 3.3.1, and shall end at a sufficient distance from exposed conductors to insure against shorting or arcing between the conductor and the shielding. The shielding shall be terminated by a crimp-type ferrule, or by soldering at the point of termination of the shielding.

3.4 Identification of wiring.- Wiring shall be identified by color-coding or, if approved by the contracting officer, may be identified by other means such as marking of terminals at both ends of each lead. Color-coding for chassis wiring of electronic equipment shall conform to Standard MIL-STD-122.

3.5 Soldering.-

3.5.1 Solder.- Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.5.2 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.5.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.6 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.7 Finish, protective.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware type items such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type I surface shall be green color (olive drab), semi-gloss enamel matching a color chip provided by the procuring agency. (See 6.10).

3.8 Riveting.- Riveting shall not be used for mounting parts such as capacitors, resistors, transformers, inductors, and miniature type sockets. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1-1/2 times the rivet diameter. Rivets shall be used where practicable in preference to other means of securing aluminum or magnesium structural parts if the joint is subjected to dynamic tension loads when the

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3.8 Riveting.- (Cont'd)

equipment is exposed to specified service conditions and if the parts do not require removal when the equipment is serviced. After riveting, the joints shall be light, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.9 Self-tapping screws.- Thread-forming screws shall not be used. Thread cutting screws shall be used only if construction is improved thereby and only in places where the screws will not require loosening or removal during operation or maintenance of the equipment. In addition, their use shall be subject to approval by the contracting officer. Chips formed in through holes by thread cutting screws shall be removed.

3.10 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain parts and subassemblies securely when the equipment is subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the equipment is subjected to specified shock test. Any permanent deformation resulting from the tests shall not cause degradation of the performance of the equipment below full specification requirements. Where practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such a manner that failure of a single rivet or screw will not free the part completely.

3.11 Tropicalization.-

3.11.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.11.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.11.2 Treatment of materials.-

3.11.2.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.11.2.2 Toxicity.- Treatment of material shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

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3.11.2.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11.2.4 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment are treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification. (See 6.2(j)).

3.11.3 Materials not resistant to moisture and fungi.-

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, on wood-flour as a filler or base.

3.12 Welding.- Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed noncritical structures. Wherever practicable, the welded joint shall conform to Standard MIL-STD-22, and shall be such that grinding on the finished weld will be unnecessary.

3.12.1 Cleaning.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.12.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.13 Marking, general.-

3.13.1 Marking.- Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group I.

3.13.2 Visibility.- Wherever practicable, parts and assemblies shall be so that their identification markings will be readily visible with minimum disassembly of the equipment.

3.13.2.1 Serial numbers.- Amplifier-Power Supply AM-598/U shall be serially numbered.

3.14 Controls.- Controls shall maintain their setting when the equipment is subjected to specified service conditions.

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3.15 Interchangeability.- The contractor's procedures and methods---including engineering, production, and quality control--shall insure interchangeability of parts as specified below. However, when matched parts are permitted by the Government, they need be interchangeable only as a pair or set.

3.15.1 Corresponding parts.- "Corresponding parts" as used herein, in connection with the interchangeability requirements, means all possible parts meeting a set of limiting values (dimensions, tolerances, ratings and limits). Such values shall be those specified in the applicable Government documents, supplemented as necessary with values established by the manufacturer of the part; however, if no Government document applies, the limiting values shall be those established by the manufacturer of the part.

3.15.2 Interchangeability requirement.- Any part shall be replaceable with any corresponding part without degrading electrical, mechanical or physical characteristics and performance below specified values and without modification of the replacement part or of any other item with which the replacement part is used. It is permissible, however, to use normal adjustments in meeting this requirement. (The foregoing requirement necessarily precludes reliance on selective assembly to meet specified requirements, or reliance on any characteristics not included in the limiting values of the part.)

3.16 Air-seal test, provision for.- The equipment enclosure shall provide means for enabling performance of the air-seal test. (See 4.5).

3.17 Suppression of radio interference.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class III. Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the unit proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.18 Electrical requirements.-

3.18.1 Primary power source.- The amplifier-power supply shall be capable of satisfactory operation from a 24 volt vehicular electrical system having a voltage range of from 22 to 30 volts d.c.

3.18.2 Output voltage.- Throughout the input voltage range of 22 to 30 volts d.c. and with the various load and temperature conditions, the output voltage shall be within the following limits:

Voltage	Min. Output Voltage d.c	Rated Output Voltage d.c.	Max. Output Voltage d.c.
B_1	55.0	60.0	65.0
B_2	117.0	130.0	137.0
A_1	1.20	1.35	1.5
A_2	1.20	1.35	1.5
C	-5.0	-5.5	-6.0

NOTES:

(a) The A_1 voltage specified in the tabulation applies for all conditions except temperature; limits in temperature tests are: 1.15 volt minimum, 1.53 maximum. The C voltage specified in the tabulation applies for all conditions except calibrate loads; limits for the calibrate loads are: 4.5 volts minimum, -6 volts maximum.

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NOTES: (Cont'd)
(b) The B_1 output voltage shall not vary more than 1.0 volt between receiver and transmitter loads for constant battery voltage.

3.18.3 No-load voltage.- No-load voltage shall not exceed the following limits with 30 volts input and all loads removed simultaneously:

B_2	145 volts d.c. (max.)
A_1	3.0 volts d.c. (max.)
A_2	2.6 volts d.c. (max.)
C	-10.0 volts d.c. (max.)

3.18.4 Input current.- The input current under the load conditions of 3.18.2 shall not exceed the following:

Load Condition	Max. Current (Amperes)
Receive	3.0
Transmit	3.5

3.18.5 Ripple.- Ripple voltage present in the output shall not exceed the following under any of the conditions of 3.18.2.

Voltage	Ripple Voltage (Volts R.M.S.)
B_1	0.02
B_2	0.05
A_1	0.005
A_2	0.005
C	0.005

3.18.6 Radio frequency noise.- Radio frequency noise from the Amplifier-Power Supply shall be suppressed to the extent necessary to prevent interference of Radio Transmitters RT-174( )/PRC-8, RT-175( )/PRC-9, and RT-176( )/PRC-10, over the frequency range from 21 to 54 megacycles when tested in accordance with paragraph 4.18. The signal-plus-noise to noise ratio of the radio set when operated from the power supply shall show no measurable difference between that of the radio set when it is operated from dry batteries having the following standard d.c. voltages:

A = 1.4 volts

B_1 = 65.0 volts

B_2 = 130.0 volts

C = 6.0 volts

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3.19 Audio output.- The maximum audio output shall be not less than 1.0 watt with 2.0 milliwatts input at 1000 cps.

3.19.1 Audio distortion.- The harmonic distortion shall be not greater than 7 percent. The method of test shall be in accordance with 4.22.

3.19.2 Audio response.- The audio response of the amplifier when measured in accordance with 4.22.1 shall be within the following limits:

Audio frequency (cycles/sec)	Audio response
1000	0 db = 1/4 watt (reference)
150	+0.5 db and -1.5 db
400	+1.0 db and -0.5 db
2500	0 db and -2.0 db
5000	0 db and -3.0 db

3.20 Operation.- Amplifer-Power Supply AM-598/U shall be subjected to the operational test of 4.25, to determine compliance with specification requirements.

3.21 Service conditions.- The equipment shall meet the following service conditions:

3.21.1 Equipment nonoperating.- Except as stated in 3.21.1(b), the equipment shall comply with the operational requirements of 3.21.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature.- Exposure in the range of +160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time. (See 4.15, 4.16 and 4.17).

(b) Relative humidity.- Relative humidity up to 97 percent for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.21.2.

(c) Elevation.- Elevation up to 25,000 feet above sea level.

(d) Immersion.- Three feet of water for 2 hours.

(e) Vibration, bounce and shock.- As described in 3.22.

3.21.2 Equipment operating.- After a warm-up period of 1 hour, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Operation.- The power supply shall be capable of operating for a period of 1000 hours without failure except for tubes and vibrator. The vibrator shall provide a minimum of 500 hours of trouble-free service when tested as specified in 4.19.

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3.21.2 Equipment operating.-

(b) Temperature.- Ambient temperature in the range of +150°F to -65°F. (The 150°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours. (See 4.15 and 4.17).

(c) Relative humidity.- Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours. (See 4.7).

(d) Elevation.- Elevation up to 10,000 feet above sea level.

3.22 Vibration, bounce and shock.-

3.22.1 Vibration.- Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.22.2 Bounce and shock.- The equipment shall meet the requirements of Table I.

Table I - Bounce and shock

Requirement	Inspection paragraph	Performance after inspection
Bounce, light equipment	4.10.3	Full specification performance. No physical damage except surface abrasion.
Shock, ballistic	4.10.1	Components shall be operable. Minor mechanical damage is permissible.
Shock, bench handling	4.10.2	Full specification performance. No physical damage.

3.23 Preconditioning.- Equipment shall be capable of meeting the inspection of Section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4. (Also see 4.3(d)).

3.24 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.2(e)(1).)

3.25 Technical literature, tools, and running spare parts.- Technical literature, tools and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the unit furnished on the order. (See 6.2).

3.26 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.1 Construction.
3.2 Wiring and cabling.
3.2.1 Slack in wires and cables.
3.2.2 Protection for wires and cables.
3.2.4 Clearance in wiring and cabling.
3.2.5 Splicing of wires.

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3.26 Workmanship.- (Cont'd)

3.2.6 Connections, general.
3.3 Grounding.
3.3.2 Shielding on wire and cable.
3.5.2 Flux and cleaning agents.
3.5.3 Process of soldering.
3.6 Cleaning of equipment.
3.7 Finish, protective.
3.8 Riveting.
3.9 Self-tapping screws.
3.10 Securing of parts.
3.11 Tropicalization.
3.12 Process of welding.
3.13.1 Marking.
3.16 Controls.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection; responsibility and classification of.-

4.1.1 Contractor responsiblity.- Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.26).

(1) Procurement inspection of items before preparation for delivery. (See 4.3).

(2) Procurement inspection of preparation for delivery. (See 4.26 and section 5.)

4.2 Preproduction inspection.- This inspection shall consist of the preproduction inspection specified in table II and the group A, group B, and group C inspection specified in Tables III, IV and V, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, bench-handling, (4) shock, ballistic, and (5) immersion. Other preproduction inspection may precede, follow, or be interspersed between the foregoing.

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Table II Preproduction inspection

Inspection	Rqt Par	Insp. Par
Altitude	3.21.1(c)	4.13
Moisture-Resistance	3.21	4.7
Vibration	3.22.1	4.12
Bounce	3.22.2	4.4, 4.10.3
Shock, ballistic	3.22.2	4.10.1
Shock, bench-handling	3.22.2	4.10.2

NOTE: The shock and bounce tests shall be performed with a load simulating Radio Set AN/PRC-8, -9 or -10 mounted in place on Amplifier-Power Supply AM-598( )/U.

4.3 Procurement inspection of items before preparation for delivery.- The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its descretion--may perform all or any part of the inspection specified in (a) to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. (Also see 6.9).

(a) Procurement inspection shall consist of group A, group B and group C inspection as specified in 4.3.1 through 4.3.3.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.23).

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4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal test (4.5) shall be next to last and the operational inspection (4.25) shall be last. (See 6.4).

Table III - Group A Inspection

Inspection	Rqt Par	Test Par	AQL Major	AQL Minor
Visual and mechanical	3.26	4.24		
Amplifier-Power Supply AM-598( )/U		4.24	1.5 dphu*	6.5 dphu
Mounting MT-759( )/PRC		4.24	1.0%	4.0%
Case, Accessories CY-1210		4.24	1.0%	4.0%
Cable Assembly, Special Purpose, Electrical CX-2031( )/U		4.24	1.0%	4.0%
Electrical				
Cable Assembly			1.0%	**
Output Voltage	3.18.2	4.14)		
Input Current	3.18.4	4.20)		
Ripple	3.18.5	4.21)	1.5%	
Audio Output	3.19	4.22.1)		
Audio Distortion	3.19.1	4.22)		
Audio Response	3.19.2	4.22.1)		
Air-Seal	3.16	4.5	1.0%	
Operational	3.20	4.25	1.0%	

* Defects per hundred units.
** All electrical defects are considered major.

4.3.2 Group B inspection.- This inspection (including sampling) shall conform to table IV and "Sampling for Expensive Testing by Attributes" in Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in Table IV, shall be as follows:

AQL	Inspection Level for Normal Inspection	Inspection Level for Reduced Inspection
6.5%	L-7	L-5

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4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table IV - Group B inspection

Inspection	Rqt Par	Insp Par	AQL (See 4.3.2.1)
Radio frequency noise measurement	3.18.6	4.18	6.5%
Immersion	3.21.1	4.9	6.5%
Interchangeability (dimension)	3.15	4.23	6.5%

4.3.2.3 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.3.3 Group C inspection.- This inspection shall be as listed in Table V.

Table V - Group C inspection

Inspection	Rqt Par	Insp Par
1000-hour operational test	3.21.2 (a)	4.19
Temperature	3.21	4.15, 4.16, 4.17
Suppression	3.17	4.11

4.3.3.1 Sampling for inspection of equipment.- Two Amplifier-Power Supplies for each group C inspection, except suppression, shall be selected each month, without regard to their quality. Sampling of equipment for suppression shall be in accordance with MIL-I-11748, for Class III.

4.3.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the Contracting Officer.

4.3.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

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4.3.5 Disposition of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Bounce preconditioning.- The unit, with shock mounts removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skanesteles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Air-seal test.- The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.6 Test conditions.- Unless otherwise specified, the following test conditions shall apply:

4.6.1 Load resistors used in electrical tests.- In order to simulate actual operation of the power supply for purposes of electrical testing and measurement, load resistors which are equivalent to and representative of the radio sets for which the power supply is designed shall be used in place of the radio sets themselves. The output loads shall be as follows:

Operation	Resistance (Ohms)				
	B_2	B_1	A_1	A_2	C
Receive Normal	Inf.	3920	2.215	-----	1000 Ohm voltmeter
Receive Squelch	Inf.	3310	2.03	-----	on
Receive Calibrate	92900	3800	2.215	5.87	10 volt
Receive Squelch Calibrate	92900	3220	2.03	5.87	scale
Transmit Normal	2800	6900	3.29	-----	14.8
Transmit Squelch	2800	5210	2.90	-----	14.8

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4.6.1 Load resistors used in electrical test.- (Cont'd)

Operation	Resistance (Ohms)				
	B_2	B_1	A_1	A_2	C
Receive Normal	Inf.	3920	2.455	----	110,000
Receive Squelch	Inf.	3310	2.25	----	110,000
Receive Calibrate	92900	3800	2.455	7.5	110,000
Receive Squelch Calibrate	92900	3220	2.25	7.5	110,000
Transmit Normal	2800	6900	2.37	----	14,800
Transmit Squelch	2800	5210	2.18	----	14,800

4.6.2 Audio amplifier tests, general.- All audio amplifier tests shall be made with the power supply operating from a 26 volt d.c. input with a resistive load (see 4.6.1) representing any one of the radio sets for which the power supply is designed.

4.7 Moisture-resistance test for equipment.- The equipment shall be tested as follows:

4.7.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the unit enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.7.2 Test method.-

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.8 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for fifteen 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.8 shall be made during the periods shown on the standard and shall comply with 4.8.

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(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance of those measurements specified in 4.8.

4.8 Performance.- Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurement	Performance during cycling
4.21	Ripple voltage	Full specification requirements
4.22	Audio distortion	Full specification requirements
4.21.1	Audio response	Full specification requirements

4.8.1 Failure.- If any equipment fails to meet the performance specified in 4.8 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.8 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.7.2(e), it does not pass the test.

4.9 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.10 Bounce and shock.-

4.10.1 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment, including shock mounts, shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.10.2 Shock test; bench-handling.- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

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4.10.2 Shock test; bench-handling.-

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.10.3 Bounce test.- The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.11 Radio interference suppression.- Inspection for suppression shall be performed in accordance with Specification MIL-I-11748, for Class III equipment.

4.12 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.21.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.12(d).)

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.13 Altitude test.- The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure, and humidity, and the performance measured. The pressure shall then be reduced to simulate 10,000 feet above sea level and stabilized for 2 hours. The performance of the equipment shall again be measured. The pressure shall then be lowered to simulate 25,000 feet above sea level and stabilized for another 2 hour period. The pressure shall then be increased to sea level conditions and the performance measured again. The equipment shall have suffered no degradation in performance from that obtained prior to this test.

4.14 Voltage output.- The power supply shall be tested for voltage output by means of a voltmeter with a resistance of 1000 ohms per volt and an accuracy tolerance of not less than one (1) percent. The unit shall be tested with input voltages of 22 and 30 volts d. c. and with the resistive loads specified in 4.6.1. The power supply shall meet the requirements of 3.18.2.

4.15 Low temperature test.- The power supply shall be placed in a chamber for three days at ambient temperature of -80°F. The temperature shall then be increased to -40°F and stabilized for a period of one hour. The unit shall be operated at this temperature with the resistive loads specified in 4.6.1. for a period of five hours and shall meet the requirements of 3.18.

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4.16 Temperature range test.- After completion of the low temperature test (4.15) operate the power supply in a chamber at an ambient temperature of -40°F for four hours, with an input voltage of 30 volts d-c and loads as specified in 4.6.1. Then increase the ambient temperature within the chamber at a rate of approximately 16°F (10° C) every 20 minutes until the temperature reaches 122°F (50°C). During this test, the power supply shall meet the requirements of 3.18.3, 3.18.4 and 3.18.6.

4.17 High temperature test.- The power supply shall be placed in a chamber for three days at an ambient temperature of + 160 F. The temperature shall then be reduced to + 132°F and stabilized for a period of one hour. The unit shall then be operated at this temperature with the resistive loads specified in 4.6.1 for a period of five hours and shall meet the requirements of 3.18.

4.18 Radio frequency noise test.- The tests for noise measurement shall be conducted with the power supply operating each of the radio receiver-transmitters listed below. With a power supply input of 24 volts d. c., measurements shall be made of the signal-plus-noise to noise ratio of the radio receivers taken as the ratio of audio output with standard modulated input to the audio output with unmodulated carrier. The ratio shall be measured at the following radio frequencies and under the following test conditions:

Receiver-Transmitter	Frequency, megacycles
RT-174( )/PRC-8	21, 24, and 27
RT-175( )/PRC-9	28, 33, and 38
RT-176( )/PRC-10	39, 47, and 54

Dummy Antenna	33 ohms in series with the signal generator
RF Input	0.5 microvolts, 15 KC deviation at 1000 cycles per second
Audio Output	1.0 milliwatt from radio receiver
Load Impedance	600 ohms

The test shall be run at room temperature and normal atmospheric conditions. The RF Signal Generator Impedance shall be 50 ohms or it shall be so adjusted and the proper calibration applied. The signal generator output shall be connected to the auxiliary antenna receptacle of the receiver transmitter.

4.19 1000-hour operational test.- The power supply shall be operated with the resistive loads as specified in 4.6.1 for a period of 1000 hours under the following conditions and shall meet the requirements 3.21.2(a).

Load Cycle: 9 minutes Receive; 1 minute Transmit

Operation: Continuous except the power supply shall be off for 10 minutes of each 24 hour period of operation.

Input Voltage:

30 volts d. c. for first 250 hours
24 volts d. c. for next 250 hours
30 volts d. c. for next 250 hours
24 volts d. c. for next 250 hours

Starting: At least once every 24 hours of the life test, the power supply shall be tested for starting with input voltages of 32 and 19.5 volts d. c. At each input voltage the power supply shall be turned on and off 5 times.

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4.19 1000-hour operational test.-(Cont'd)

Starting: At least once every 24 hours of the life test, during those days that the contractor's plant is in operation, the power supply shall be tested for starting with input voltages of 32 and 19.5 volts d-c. At each input voltage the power supply shall be turned on and off 5 times.

4.20 Input voltage range.- The power supply shall be tested for operation with input voltage of 22 and 30 volts d. c. At each of the input voltages, the power supply shall be switched on and off at least five (5) times. The power supply shall not fail to start and there shall be no evidence of failure of any part of the equipment During this test the power supply shall be connected to a resistive load. (See 4.6.1).

4.21 Ripple voltage.- With inputs of 22 and 30 volts d. c., the power supply shall be tested for ripple voltage in the output circuit by means of an electronic voltmeter such as model 300 manufactured by Ballantine Laboratory, Incorporated, Boonton, New Jersey, or equal. The unit shall be operated with a resistive load (paragraph 4.6.1) representing any one of the radio sets for which the power supply is designed, and it shall meet the requirements of 3.18.5.

4.22 Audio distortion.- The input signal to the amplifier shall be adjusted to obtain 250 milliwatts output at frequencies of 150, 400, 1000, 2500, and 5000 cycles. The distortion output of the amplifier shall comply with the requirements of 3.19.1

4.22.1 Audio response.- The input signal to the amplifier shall be adjusted to obtain 250 milliwatts output at a frequency of 1000 cycles. With this input signal the audio frequency response shall be measured at frequencies of 150, 400, 1000, and 5000 cycles and shall be in accordance with 3.19.2.

4.23 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

4.24 Visual and mechanical inspection of equipment.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.25 Operational inspection.- A voice modulated signal shall be received from a RT-174, RT-175 or RT-176 Transmitter. The transmitter shall be separated from the receiver by a distance of at least 50 feet. The equipment shall be powered by Amplifier-Power Supply AM-598/U. Reception and transmission qualities shall indicate proper operation of the system.

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4.26 Procurement inspection of preparation for delivery.- The minimum inspection performed by the contractor shall be as specified in 4.26.1 and 4.26.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.26.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on Table VI.

Table VI

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Conforming wraps are not snug-fitting and contain voids.
4. Cushioning or padding omitted.	4. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect or illegible.	
10. Different stock numbered items in the same unit package.	

4.26.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to MIL-STD-105. Inspection Level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on Table VII.

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TABLE VII
PACKING, AND MARKING THEREOF
MAJOR

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Box closure not as specified.
4. Type, grade, class, and style of the shipping container not as specified.
5. Strapping omitted.
6. Strapping inadequate or incorrectly applied.
7. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
8. Shipping documents or packing list omitted.
9. Stock number omitted, incorrect, or illegible.
10. Nomenclature omitted, incorrect, or illegible.
11. Marking of quantity of items in pack omitted, incorrect, or illegible.
12. Destination marking omitted, incorrect, or illegible.
13. Service designation (color marking) omitted.
14. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
15. Overseas code marking omitted, incorrect, or illegible.

TABLE VII
PACKING, AND MARKING THEREOF
MINOR

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 9 thru 15 listed under major defects omitted, incorrect, or illegible.

4.26.3 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.26.4 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.27 Rough handling test (preparation for delivery).- When rough handling test is required by the contract (see 6.2(e), functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling, and shall meet the requirement of 4.25.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Amplifier-Power Supply AM-598( )/U shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting Type I Class b bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Spare tubes, vibrator, fuses.- No packaging required.

5.1.1.3 Stowage within Case CY-1210( )/U.- Stow the items specified in 5.1.1.2 within the case. Close the case and secure all fastenings.

5.1.1.4 Case CY-1210( )/U with contents.- Each case with contents shall be packaged Method III as follows: Inclose each case within a wrap of Type II single-faced, flexible corrugated fiberboard conforming to Specification LLL-F-291. Secure the wrap with Class 2 gummed paper tape conforming to Specification UU-T-111.

5.1.1.5 Hardware kit.- The elements of each hardware kit shall be packaged Method III as follows: Place the elements of each hardware kit within a cotton mailing bag conforming to Specification DDD-B-20. Close the bag by means of the drawstring.

5.1.1.6 Cable Assembly CX-2031/U.- Each cable assembly shall be packaged Method III as follows: Wind the cable assembly into a coil of smallest possible dimensions and tie at 3 places with 6 ply cotton twine conforming to Specification T-T-871, securing the free ends of the assembly to the coil with the ties. Wrap the coiled assembly within Type I, single-faced flexible corrugated fiberboard conforming to Specification LLL-F-291. Secure the cushioning with Class 1 gummed paper tape conforming to Specification UU-T-111.

5.1.1.7 Consolidated package.- Consolidate the items packaged as specified in 5.1.1.3 through 5.1.1.6 together within a regular slotted style corrugated fiber box Class 1 W5c conforming to Specification PPP-B-636. Fill all voids with pads fabricated of single-faced flexible corrugated fiberboard. Box closure shall be in conformance with the appendix of the referenced specification.

5.1.1.8 Amplifier-Power Supply AM-598( )/U, with Mounting MT-759( )/PRC.- Position and secure the amplifier upon the mounting. Each amplifier with mounting shall be packaged Method IC-5 as follows: Cushion each amplifier on all surfaces with cells or pads or both, fabricated of double-faced corrugated fiberboard having a minimum dry bursting strength of 200 pounds, and designed to absorb the shock of impact in handling and transit. Cells and pads shall have water resistant qualities equal to the box in which they will be placed. Place the cushioned amplifier, together with the consolidated package (see 5.1.1.7) within a regular slotted style corrugated fiber box, Class 2 W5c, conforming to Specification PPP-B-636. Fill all voids with cells or pads fabricated of double-faced corrugated fiberboard to prevent independent movement of the box contents. Box closure shall be in accordance with the appendix of the referenced box specification. In addition seal all seams and joints with Type III, Class 1, olive-drab pressure sensitive waterproof tape conforming to Specification PPP-T-60.

5.1.2 Level C.- Amplifier-Power Supply AM-598( )/U shall be packaged in accordance with commercial practice, and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

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5.2 Packing.-

5.2.1 Level A.- Amplifier-Power Supply AM-598( )/U packaged as specified in 5.1.1.8 shall be packed for shipment, one each, within a close-fitting style A, Type I, Class 2, overseas type boxes, wood, cleated-plywood, conforming to Specification PPP-B-601, or Class 2, style 4, boxes, wood, nailed and lockcorner, conforming to Specification PPP-B-621. Place the technical manual (see 5.1.1.1) between the contents and lid of the box. Box closure shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification, only for direct shipment to ports.

5.2.2 Level B.- Amplifier-Power Supply AM-598( )/U shall be packed for shipment as specified in 5.2.1 except boxes shall be domestic class or type as applicable.

5.2.3 Level C.- Amplifier-Power Supply AM-598( )/U shall be packed for shipment in a manner conforming to the requirments of Uniform Freight Classification No. 4, and Consolidated Freight Classification, Rule 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed, at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The Amplifier-Power Supply AM-598( )/U is used to furnish the d. c. power required for the operation of the Receiver-Transmitters, RT-174( )/PRC-8, RT-175( )/PRC-9 and RT-176( )/PRC-10, and, in addition, the audio amplifier which it contains permits the radio set receiver to operate a 600-ohm loudspeaker.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.24).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

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6.2 Ordering data.- (Cont'd)

d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. This data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (3.25).

(i) Quantity of tools and running spare parts required. (See 3.25).

(j) Submission of the statement of treatment referenced in 3.11.2.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AM-598W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Location of operational inspection and air-seal test.- It is desirable that the operational inspection (4.25) and the air-seal test (4.5) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.2.)

6.6 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.7.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

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6.7.2 Testing.- Testing consists of determination of compliance, using technical means.

6.8 Equipment division.- The terms used herein for equipment divisions conform to Standard MIL-STD-280.

6.9 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.10 Color.- The color chip furnished by the procuring agency will match color chip No. 2430 of superseded Specification TT-C-595, and may be obtained upon request to Commanding General, U. S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania, ATTN: SIGSU-J4b. This color chip does not match any color chip of Federal Standard Number 595.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsiblity nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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